



BEFORE

By 1850, the age of sail was waning as steam engines increasingly replaced ships using wind and canvas.

FIRST IRONCLADS

The Crimean War (1853–56) augured **the end of the wooden hull** for warships, as powerful new rifled artillery and explosive shells encouraged navies to protect their vessels with iron armor. By 1861, the **first steam-powered ironclad warships**, the French *Gloire* and British *Warrior*, were roving the seas. The **first combat between ironclads** was the engagement, on March 9, 1862, between the Union's *Monitor* and the Confederacy's *Virginia* (or *Merrimack*) (see [The Battle of Hampton Roads](#)).

MINES AND SUBMARINES

Torpedoes—floating kegs of gunpowder also known as mines—**first appeared** during the American Revolution. They were deployed in large numbers during the Crimean War.

Since the end of the 18th century, many navies had shown a passing interest in **building submersible warships**, but most experiments were not promising. With the outbreak of the Civil War, efforts were quickly renewed as both **Union and Confederate engineers raced** to overcome design challenges and develop the **first dependable and operationally effective submarine**.

When war broke out, the U.S. Navy had fewer than 70 serviceable ships, while the Confederates counted less than a dozen. Yet “inequality of numbers,” the Southern Secretary of the Navy, Stephen Mallory, asserted, “may be compensated by invulnerability.” He was referring to the ironclads. He hoped that they would prove so impervious to enemy guns that a single one might breach the Union blockade to attack and burn New York City.

Ironclads and mines

In the event, the South’s lack of industrial resources forced Mallory to choose the simplest ironclad designs. In contrast, Lincoln’s stern-faced navy secretary, Gideon Welles, harnessed factories, foundries, and shipyards to build technologically sophisticated ironclads, 65 in all. Most ships were refinements on the first Union ironclad, the *Monitor*, launched in 1862, but were longer, mounted with heavier guns, and even had more than one turret. Low-slung and formidable, they were more versatile than any of their Confederate counterparts. Mallory had another card to play: “torpedoes.” The Confederacy’s secretive Torpedo Corps and Submarine Battery Service experimented with all kinds of these “infernal machines.”